JYOSHNA NALUVALA

Business Data Analyst | SQL • Power BI • Stakeholder Management

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PROFESSIONAL SUMMARY

Business Data Analyst with 3+ years of experience across insurance, healthcare claims, and higher-education analytics, specializing in KPI reporting, dashboard development, business performance tracking, and stakeholder-ready insights. Proficient in SQL, Power BI, Tableau, and Python, with proven ability to translate business requirements into actionable reporting. Reduced post-release dashboard defects by 30%, expanded data validation coverage by 20%, and automated 25% of manual reporting workflows across enterprise, healthcare and academic environments.

TECHNICAL SKILLS

Programming & Querying: Advanced SQL (Joins, window functions, CTEs, query optimization), Python (Pandas), Snowpark.
Business Intelligence & Visualization: Power BI (DAX, Power Query), Tableau, MicroStrategy, Excel, Dashboard Development, Executive Reporting, KPI Reporting, Data Storytelling.
Data Modeling & Warehousing: Snowflake, dbt, BigQuery, Teradata, Vertica, SQL Server, PostgreSQL, Star Schema, Dimensional Modeling, Source-to-Target Mapping.
Data Quality & Validation: dbt Tests, Data Reconciliation, Query Validation, Data Integrity, Reporting Accuracy Validation.
Analytics & ML: A/B Testing, Hypothesis Testing, Regression, Predictive Modeling, Forecasting (Snowflake Cortex ML), Descriptive & Inferential Statistics, Correlation Analysis, Model Evaluation.
Data Analysis & Modeling: Data Cleaning, Exploratory Data Analysis (EDA), KPI Analysis, Root-Cause Analysis, Business Reporting, Data Pipeline Validation, Business Performance Tracking, Ad Hoc Analysis, Stakeholder Reporting.
Tools & Collaboration: JIRA, Confluence, Git/GitHub, Agile/Scrum, SDLC, UAT, Requirements Gathering, Stakeholder Management, BRD & Data Dictionary Documentation.

EXPERIENCE

Research Data Analyst — **Aug 2024 - Present**
Texas A&M University — Corpus Christi, Texas

- Validated and standardized 15+ enrollment and research datasets (~400K records) across Snowflake and BigQuery using SQL/Python, cutting data errors in institutional reports supporting $2M in grant submissions.
- Built 5 Tableau dashboards and Excel reporting models using SQL CTEs and window functions, enabling leadership and faculty to track enrollment trends and departmental metrics without manual pulls and cutting manual reporting time by 25%.
- Standardized metadata and data dictionaries across 15-20 datasets spanning cloud and relational environments, improving data reuse readiness for institutional and grant reporting cycles.
- Built a library of reusable SQL validation queries and Excel reconciliation trackers to catch data gaps, transformation errors, and source-to-report mismatches replacing ad hoc manual checks with a repeatable, scalable validation workflow.

Data Analyst — **Apr 2023 - Aug 2024**
MassMutual — Hyderabad, India

- Validated 300+ KPIs across enterprise MicroStrategy dashboards using SQL, enforcing business rules, maintaining data integrity, and ensuring reporting accuracy for stakeholder-facing insurance reports.
- Designed and executed source-to-target reconciliation checks across enterprise insurance datasets validating record counts, field transformations, duplicates, and calculation logic expanding validation coverage by 20%.
- Validated MicroStrategy dashboard KPIs, filters, calculations, aggregations, and drill-down logic to ensure reporting accuracy and consistency, contributing to a 30% reduction in post-release reporting defects.
- Investigated source-to-dashboard mismatches to identify root causes, then documented defect details, business impact, and resolution status in JIRA and Confluence improving cross-team communication between QA, engineering, and business stakeholders.
- Partnered with business stakeholders to standardize KPI definitions, validation rules, and reporting sign-off criteria, improving consistency of enterprise insurance dashboards and reducing repeat clarification cycles during reporting releases.

Junior Data Analyst **May 2021 - Mar 2023**
FHPL-Insurance TPA Hyderabad, India

- Built 7 Power BI dashboards using SQL, PostgreSQL/SQL Server, and Excel to monitor ~7K healthcare claims per month, denial trends, pending claims, and hospital-wise TAT/SLA performance for operations leadership.
- Designed source-to-target mappings, data dictionaries, and reporting definitions for key claims fields (claim ID, member ID, policy number, hospital code, claim status, settlement amount), improving consistency across recurring healthcare operations reports.
- Validated and reconciled claims data across operational systems, SQL reporting tables, and Excel trackers, identifying duplicate records, missing fields, incorrect claim statuses, date mismatches, and settlement discrepancies, reducing manual reconciliation effort by 25%.
- Supported SDLC activities for a legacy Teradata decommissioning initiative by gathering reporting requirements, documenting source-to-target mappings, validating business rules, and supporting migration of claims/reporting datasets to Vertica-based cloud reporting environments.
- Collaborated with data engineering, QA, and business operations teams across SDLC phases, including requirements gathering, development validation, UAT, production cutover, and post-migration issue resolution, reducing dependency on legacy Teradata reporting.

PROJECTS

Policy Revenue Leakage & Reconciliation System | Snowflake, dbt, SQL, Power BI

- Architected an end-to-end insurance analytics mart in Snowflake using dbt (staging → intermediate → mart layers), transforming 6 linked datasets (customers, agents, policies, premiums, claims, adjustments) into a tested, BI-ready star schema.
- Implemented automated dbt data-quality tests (uniqueness, referential integrity, accepted values, and business-rule validation) across the full pipeline to catch data issues before they reached reporting.
- Designed claims leakage detection logic identifying duplicate claims (5.8% of claims), claims filed after policy expiration (7.1%), and high-severity outlier claims (4.3%), assigning each a leakage risk score for investigation prioritization.
- Built a Power BI control tower dashboard surfacing loss ratio, premium trends, agent performance, and a data quality reconciliation scorecard for stakeholder reporting.

Healthcare Claims Risk & Predictive Intelligence Platform | Snowflake, dbt, Snowpark, Cortex ML, Power BI

- Built a 15-model dbt pipeline in Snowflake on 558K+ real Medicare-derived healthcare claims (CMS Kaggle), passing 22 of 23 automated data-quality tests spanning referential integrity, accepted values, and statistical outlier detection.
- Trained a Snowpark logistic regression classifier on provider-level claims features to predict potential fraud, achieving 0.91 ROC AUC and 87% recall on a held-out test set for identifying claim volume and reimbursement variability as the strongest fraud predictors.
- Applied Snowflake Cortex ML native forecasting to predict daily claims reimbursement trends, evaluating model accuracy and calibration across daily vs. weekly aggregation grains to select the best-performing configuration.
- Built 5 additional analytical marts surfacing physician network risk, beneficiary population health segmentation, and diagnosis-level cost drivers crosscut against provider fraud status.

EDUCATION

Master of Science in Data Science, Texas A&M University – Corpus Christi Aug 2024 –May 2026
Notable Courses: Data Analysis, Database Management, Machine Learning, Statistical Modeling, Data Visualization.
Bachelor of Science in Mathematics, Statistics and Computer Science – KU Warangal 2019 - 2022
Notable Courses: Python Programming, Statistics, Mathematics, Programming, Database Systems, Probability, Data Structures

CERTIFICATIONS

Microsoft Certified: Power BI Data Analyst Associate (PL-300)
Data modeling, Power Query transformations, DAX calculations, report design, and interactive dashboard development.
SnowPro® Core Certification - Snowflake
Cloud data warehousing, SQL performance, data loading, security, and Snowflake architecture fundamentals.
Microsoft Certified: Fabric Analytics Engineer Associate (DP-600)
Lakehouse architecture, semantic modeling, Power BI integration, and analytics engineering with Microsoft Fabric.